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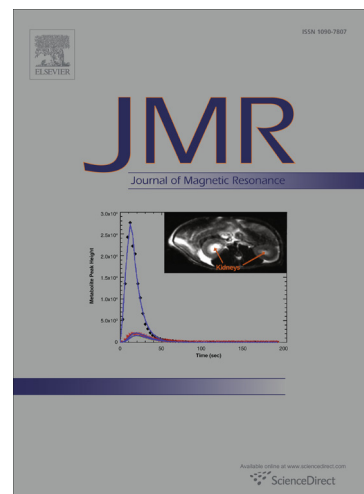
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Systematic bias in NMR diffusion measurements on polydisperse systems

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Abstract

Least-squares fitting of the Stejskal-Tanner equation is a routine process in the measurement of molecular diffusion coefficient (MDC) using Nuclear Magnetic Resonance (NMR) Spectroscopy. It is simple and elegant. However, a bias of the MDC is noticed when the system is polydisperse. This is due to improper accounts of the diffusion coefficient distribution. Eventually, it leads to a discrepancy between the observed MDC and the statistical mean value of the distribution. To reveal the discrepancy, an analytical solution is derived when the diffusion data is taken a logarithmic linearization. Computer simulation is also applied to obtain a non-linear regression result. For a Gaussian distribution of the MDCs, the bias is proportional to the square of the distribution width (linear regression), but it is also inversely proportional to the statistical mean value of the distribution (non-linear regression). This indicates that the MDC derived from Stejskal-Tanner equation only holds well for narrow distribution of MDCs. Otherwise, molecular radius derived from the Stokes-Einstein equation needs to be reconsidered due to the incorrect estimation of the MDC.

Keywords: molecular diffusion coefficient, least-squares fitting, diffusion coefficient distribution, non-linear regression;

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Introduction

Molecular diffusion coefficient (MDC), according to the Stokes–Einstein equation,^[1] correlates the random Brownian motion with the molecular size (radius). Tremendous applications have sprung out since the first NMR measurement of MDC by Stejskal and Tanner in 1965.^[2] They include monitoring the process of protein denaturation,^[3] protein and ligand binding,^[4–6] chemical reaction,^[7] medicine development,^[8] etc. MDC measurement is particularly useful in sorting out various molecules in compounds, such as polymers,^[9,10] human body fluids (blood, urine, and cerebrospinal fluid),^[11,12] wines,^[13] liquid crystals,^[14] etc. Technically, considerable attentions have been paid to improve the NMR diffusion experiments for measuring slower MDCs,^[15,16] increasing spectral resolution by multidimensional technique^[9,10,17] and quantitative determination of MDCs.^[18–21]

In practice, a set of diffusion data are collected as a function of the square of the pulsed field gradient (PFG), leading to a diffusion profile.^[2] The diffusion coefficient can be determined by least-squares fitting of the Stejskal–Tanner equation^[2] to diffusion data where only a single MDC value is assumed. This is a good approximation for small molecules. However, much more attentions are usually paid to larger and complicated molecules, such as polymers and bio-macromolecules. Because of different structures, chain lengths, or molecular weights, the diffusion coefficients of these molecules are always not the same but

distributed around the statistical mean value. In these cases, people still use the same fitting strategy, taking for granted that the statistical mean value of the distribution could be derived. Unfortunately, a systematic bias is introduced in this method which can be large if the distribution is broad. This probably would lead to a wrong estimation of the molecular size, especially for polymers and bio-macromolecules.

In this paper, an analytical solution of the Stejskal-Tanner equation is derived to account for the discrepancy. The result is then extended to a non-linear least-squares fitting. The MDC bias exists in both linear and non-linear fitting results, showing the phenomenon appearing disregarding the methods utilized. Computer simulation based on a Gaussian model agrees well with the theoretical prediction.

Theory

According to the Stokes–Einstein equation, if a molecule can be approximately considered in a sphere shape, the diffusion coefficient (D) of the molecule in solution can be expressed by

$$D = \frac{k_B T}{6\pi\eta r}, \quad (1)$$

where k_B is the Boltzmann constant, T is the temperature in Kelvin, η is the viscosity and r is the radius of the molecule. For small molecules of the same species in

ideally homogeneous solution, the MDCs can be considered of basically the same.

In this case, the Stejskal–Tanner equation well describes the signal decay in an NMR diffusion experiment

$$I_j = A \exp(-D \gamma^2 \delta^2 g_j^2 \Delta'), \quad (2)$$

where the subscript j ($j = 1, 2, \dots, n$) represents the index of Pulse Field Gradient (PFG) strengths used in the experiment, I_j and A are the signal intensities corresponding to the PFG strength (g_j) and in the absence of any gradient pulses, respectively. D is the diffusion coefficient, γ is the gyromagnetic ratio of the molecule being measured, δ is the gradient duration and Δ' is the effective diffusion delay. The experimental parameters γ , δ , g_j and Δ' are known while the intensity (A) and the diffusion coefficient (D) are left to be solved. Conventionally, a set of g values are used and least-squares regression methods, e.g. the Levenberg–Marquardt algorithm^[22], are implemented to determine A and D .

However, as mentioned above, for complicated molecules, the corresponding MDCs may not be the same. It is more likely to have a distribution of the MDCs.

Here we assume that the MDCs follow a Gaussian distribution for simplicity. Thus, the distribution function can be written as

$$h(D) = \frac{A_0}{c \sqrt{2\pi}} \exp\left(-\frac{(D - D_0)^2}{2\sigma^2}\right), \quad (3)$$

where $h(D)$ represents the distribution function of D , σ is the standard deviation, A_0 is the integrated intensity and D_0 corresponds to the statistical mean value (center) of the distribution. A schematic representation of a Gaussian distribution of MDCs is illustrated in FIG. 1, where ζ denotes the full width at half maximum (FWHM) of the distribution, which is $\zeta = \sigma\sqrt{8 \times \log 2}$.

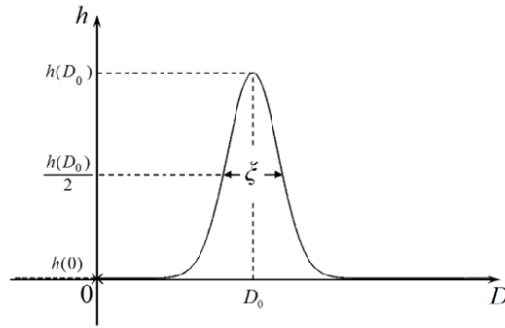


FIG. 1. Schematic representation of a normal distribution of MDCs, where D_0 is the distribution center, ζ is the full width at half maximum and the cross point occurs at $D = 0$.

Consequently, the signal intensity can be derived referring to Eqs. (2) and (3),

$$I_j = \int_0^{\infty} h(D) \exp(-Db_j) dD \approx \int_{-\infty}^{\infty} \frac{A_0}{c\sqrt{2\pi}} \exp\left(-\frac{(D-D_0)^2}{2c^2} - Db_j\right) dD, \quad (4a)$$

and the integral result of (4a) is

$$I_j = A_0 \exp\left(\frac{\sigma^2 b_j^2}{2} - D_0 b_j\right), \quad (4b)$$

where $b_j = \gamma^2 \delta^2 g_j^2 \Delta'$ is the preprocessed experimental parameter. The approximate equal mark in Eq. (4a) indicates that the integral region from $-\infty$ to 0 can be neglected if only $h(0)$ is much smaller than $h(D_0)$.

Eq. (4b), compared to the original Stejskal-Tanner equation, suggests a systematic bias between the observed diffusion coefficient (from the Stejskal-Tanner equation) and the statistical mean value of the MDCs. It is not a calculation error but a systematic deviation due to a multi-distribution of the MDCs.

To determine the character of this bias, analytical solution is deduced under the performing of logarithmic linearization on both sides of Eqs. (4b) and (2),

$$\log I_j = \log A_0 + \left(\frac{\sigma^2 b_j^2}{2} - D_0 b_j \right) = \log A - D_{app} b_j, \quad (5)$$

$$\left(\frac{\sigma^2 b_j^2}{2} - D_0 b_j \right) = (\log A - \log A_0) - D_{app} b_j. \quad (6)$$

Here we use the name “the apparent diffusion coefficient D_{app} ” to denote the diffusion coefficient obtained from the Stejskal-Tanner equation. For all experimental parameters and data ($h(0) \ll h(D_0)$), the above equation leads to

$$\begin{bmatrix} 1 & -b_1 \\ 1 & -b_2 \\ \vdots & \vdots \\ 1 & -b_n \end{bmatrix} \begin{bmatrix} \log A - \log A_0 \\ D_{app} \end{bmatrix} = \begin{bmatrix} \frac{\sigma^2 b_1^2}{2} - D_0 b_1 \\ \frac{\sigma^2 b_2^2}{2} - D_0 b_2 \\ \vdots \\ \frac{\sigma^2 b_n^2}{2} - D_0 b_n \end{bmatrix}. \quad (7)$$

By solving the above system of linear equations with respect to $\log A - \log A_0$ and D_{app} , the analytical solution becomes

$$\log A - \log A_0 = \frac{1}{2} \frac{(\sum_{j=1}^n b_j^2)^2 - \sum_{j=1}^n b_j \sum_{j=1}^n b_j^3}{n \sum_{j=1}^n b_j^2 - (\sum_{j=1}^n b_j)^2} \sigma^2, \quad (8)$$

$$D_{app} - D_0 = \frac{1}{2} \frac{\sum_{j=1}^n b_j \sum_{j=1}^n b_j^2 - n \sum_{j=1}^n b_j^3}{n \sum_{j=1}^n b_j^2 - (\sum_{j=1}^n b_j)^2} \sigma^2. \quad (9)$$

Since the relationship between the standard deviation σ and the FWHM ξ is

$\sigma = \xi / \sqrt{8 \times \log 2}$, Eqs. (8) and (9) can also be expressed by

$$\log A - \log A_0 = \mu \xi^2, \quad (10)$$

$$D_{app} - D_0 = \lambda \xi^2, \quad (11)$$

where μ and λ are two factors determined by

$$\mu = \frac{1}{16 \times \log 2} \frac{(\sum_{j=1}^n b_j^2)^2 - \sum_{j=1}^n b_j \sum_{j=1}^n b_j^3}{n \sum_{j=1}^n b_j^2 - (\sum_{j=1}^n b_j)^2}, \quad (12)$$

$$\lambda = \frac{1}{16 \times \log 2} \frac{\sum_{j=1}^n b_j \sum_{j=1}^n b_j^2 - n \sum_{j=1}^n b_j^3}{n \sum_{j=1}^n b_j^2 - (\sum_{j=1}^n b_j)^2}. \quad (13)$$

The above analytical expressions explicitly reveal the relationship between the MDC bias and the square of the FWHM (or the standard deviation). It indicates that:

- (a) a bias is introduced between the observed value and the statistical mean value of the MDCs; (b) the bias is proportional to the square of the FWHM; and (c) the proportionality factor is affected by the experimental parameters.

Nevertheless, owing to the change in statistical weight caused by taking a logarithm, this analytical solution derived by linear regression does not give the same result as non-linear regression of the experimental data to an exponential function. Besides, it is remarkable that the drawback of the linear regression becomes obvious when the experimental data are of high levels of attenuation. This is because experimental errors in very small signals have little effect on non-linear regression to an exponential, but a very large effect on linear regression. Thus the above analytical solutions are only accurate for data with low attenuation, where the change of the statistical weights of noises does not affect too much.

Computer Simulations and Discussion

To verify the above analysis results, a set of simulated diffusion data constructed of normally distributed MDCs were generated, where the distribution center (D_0) was $5.0 \times 10^{-11} \text{ m}^2/\text{s}$. The corresponding FWHMs (ξ) was $2.0 \times 10^{-11} \text{ m}^2/\text{s}$. The simulated experimental parameters were $\delta = 3 \text{ ms}$, $\Delta' = 299 \text{ ms}$, $\gamma = 2.6752 \times 10^4 \text{ G}^{-1}\text{s}^{-1}$ (the gyromagnetic ratio of proton) with 32 g_j values equally spaced from 2 to 64 Gauss/cm. The distributions and the diffusion parameters were set in this way to mimic practical situations of polymers or bio-macromolecules. For simplicity, we set the intensity $A_0 = 1$ for all simulations run in this study. Afterwards, the data was fitted by the Stejskal-Tanner equation, which is Eq. (2), using both linear and

non-linear regression method. The simulated data and the regression lines are plotted in FIG. 2.

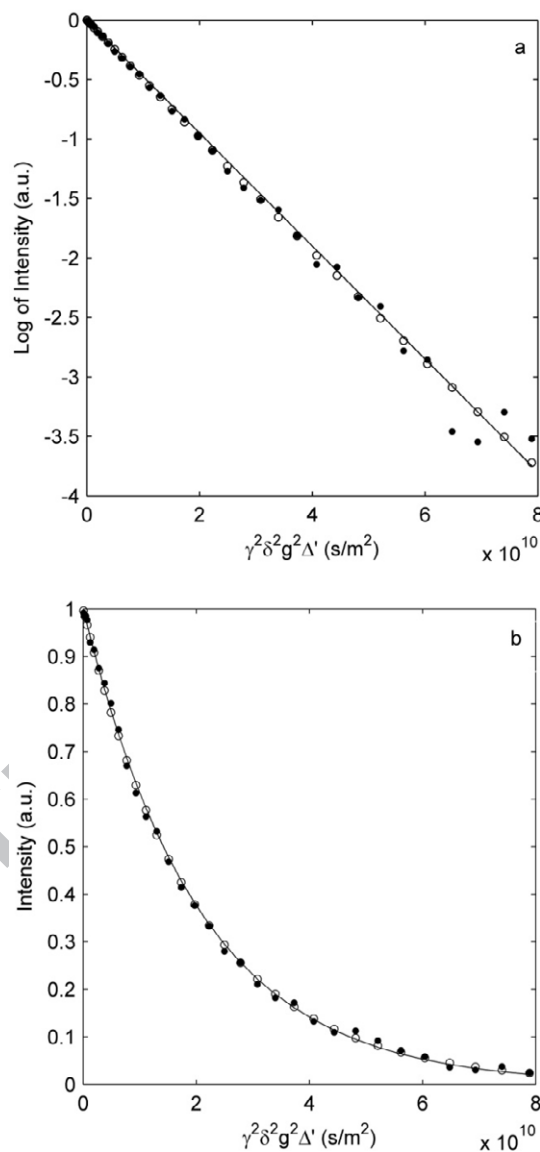


FIG. 2. A set of simulated diffusion data with center diffusion coefficient $5.0 \times 10^{-11} \text{ m}^2/\text{s}$ and FWHM $2.0 \times 10^{-11} \text{ m}^2/\text{s}$ are shown in both FIG. (a) and (b) (the open circles). The corresponding linear regression line of Eq. (2) (the solid line in FIG. 2-a) and non-linear regression line of Eq. (2) (the solid

line in FIG. 2-b) is also shown. The solid circles in both FIG. (a) and (b) are the same simulated diffusion data added with 1% Gaussian white noises, respectively.

As illustrated in FIG. 2, the regression lines represent the best fit of the data points. For linear regression fitting of Eq. (2), predictably, the obtained MDC value is deviated from the distribution center. Similarly, there also exists a deviation between the obtained MDC value and the distribution center under non-linear regression fitting. This can be confirmed from FIG. 2, which the black solid lines representing the best regression lines have a little deflection of the data. The fitted MDC of Eq. (2) is $4.7472 \times 10^{-11} \text{ m}^2/\text{s}$ in linear case, and $4.8897 \times 10^{-11} \text{ m}^2/\text{s}$ in non-linear case. The bias ($D_{app} - D_0$) is $-2.5280 \times 10^{-12} \text{ m}^2/\text{s}$ in linear case, and $-1.1034 \times 10^{-12} \text{ m}^2/\text{s}$ in non-linear case. The bias by these parameters would lead to about 2% to 5% systematic error in routine MDC measurements, which indicates a non-ignorable deviation introduced by the conventional mono-exponential fitting method.

It is worth noticing that taking logarithm of the exponential equation carries the risk of noise amplification, especially for low signal to noise ratio (SNR) data. To represent the impact of the noise, we added 1% Gaussian white noise on the original diffusion signal and took logarithm of the noise data afterwards. The black solid circles in FIG. 2(a) illustrates the effect of the growing noise by applying logarithmics, while the black solid circles in FIG. 2(b) reveals a uniform noise level on each diffusion data. Further noise tests were implemented with noise levels range

from 0.01% to 1.0%, and both linear and non-linear regression results were shown in Table 1. It says that this systematic bias can be accurately estimated under 0.1% noise level.

Table 1. Noise testing results using both linear and non-linear regression method are listed with 100 repeating times for each set of data. The set of simulated diffusion data are of center diffusion coefficient $5.0 \times 10^{-11} \text{ m}^2/\text{s}$ and FWHM $2.0 \times 10^{-11} \text{ m}^2/\text{s}$.

Noise Level (%)		0.01	0.02	0.05	0.1	0.2	0.5	1.0
$D_{\text{app}} - D_0 (\times 10^{-12} \text{ m}^2/\text{s})$	Linear	-2.5293	-2.5215	-2.5348	-2.5089	-2.5482	-2.4104	-1.8477
	Non-linear	-1.1034	-1.1021	-1.1030	-1.1044	-1.1138	-1.1138	-1.1347
R-RMS (%)	Linear	0.48	0.93	2.59	5.74	11.49	24.07	61.10
	Non-linear	0.31	0.60	1.48	3.41	7.09	15.74	35.10

These weaknesses of the logarithmics indicate that non-linear regression could be a more proper method to deal with the experimental data than the linear one, because most of the experimental data contain various noises which lead to a relatively low SNR.

As discussed above, an approximation is introduced in Eq. (4a), which indicates that the analytical solution holds only in cases where the integral from $-\infty$ to 0 is ignorable. To test the performance of this bias with FWHMs, we generated 62 sets of simulated data of decays of normally distributed MDCs using above mentioned experimental parameters. In order to keep the approximation valid, the following situation

$$R = \frac{h(0)}{h(D_0)} < 0.1\% \quad (14)$$

should be satisfied. Otherwise, there would be too many negative diffusion coefficients which cannot be ignored. Hence, all the distributions were centered at $5.0 \times 10^{-11} \text{ m}^2/\text{s}$ while equally spaced FWHMs from 1.0×10^{-12} to $3.15 \times 10^{-11} \text{ m}^2/\text{s}$ were tested. The maximum ratio of R is 0.0925%. Afterwards, the Stejskal–Tanner equation was fitted using both linear and non-linear regression method respectively. The corresponding biases are plotted versus the square of FWHMs in FIG. 3, where a predicted bias relation ($D_{app} - D_0 = -6.3199 \times 10^9 \xi^2$) (in linear case) and a regression line ($D_{app} - D_0 = -2.8149 \times 10^9 \xi^2$) (in non-linear case) depicted by solid lines are also shown.

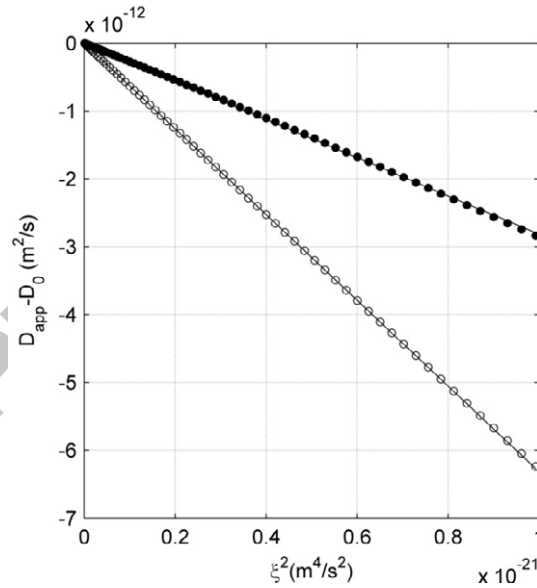


FIG. 3. Relationship between the diffusion bias and the square of the distribution width are shown. The biases are depicted by open circles (linear regression) and solid circles (non-linear regression) while the solid line represents the predicted bias relation (linear regression) and the fitted regression line (non-linear regression).

From FIG. 3, one can see that, in both linear and non-linear case, the observed diffusion coefficient bias is almost perfectly agrees with the regression line. But the bias is different between linear and non-linear regressions. This is due to the change of the statistical weight. Consequently, non-linear regression result gives a smaller deviation between the apparent MDC value and D_0 than that of the linear regression. This indicates that non-linear regression method could obtain more acceptable results than the linear one. Further increase of the FWHM, the observed MDC would have a big deviation on the statistical mean value of the distribution. This systematic bias would lead to an incorrect estimation on MDC.

Under these circumstances, logarithmetics would carry more fitting error than direct fitting using non-linear regression method. Similar to Eq. (11), we were able to derive the following relationship of the non-linear form

$$D_{app} - D_0 = \lambda' \xi^2 \quad (15)$$

Unlike the linear regression, the proportional factor λ' in non-linear form changes slightly with D_0 . The MDC distribution centers range from $5.0 \times 10^{-11} \text{ m}^2/\text{s}$ to $1.05 \times 10^{-10} \text{ m}^2/\text{s}$ with $0.5 \times 10^{-11} \text{ m}^2/\text{s}$ step length were tested, while maintaining all the other experimental parameters to be the same as preceding part of the text. The MDC biases were calculated for each D_0 and ξ . A three dimensional pseudo color graph contains both linear (surface a) and non-linear (surface b) results are shown in FIG. 4.

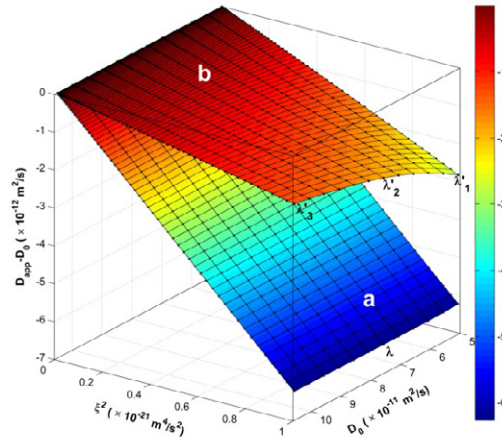


FIG. 4. Three dimensional pseudo color graph shows the relationship between the diffusion bias, the center diffusion coefficient and the square of the distribution width. The biases are depicted by different colors. Surface (a) represents the linear regression result while surface (b) is the non-linear regression result. The mark λ and the three marks (λ'_1 , λ'_2 , λ'_3) in the graph stand for the different slope of linear regression and non-linear regression results, respectively. Their values are: $\lambda'_1 = -2.81 \times 10^9$ s/m², $\lambda'_2 = -1.88 \times 10^9$ s/m², $\lambda'_3 = -1.33 \times 10^9$ s/m² and $\lambda = -6.32 \times 10^9$ s/m².

The slightly twisted pseudo color surface (surface b) suggests the dependence between the proportional factor and the center diffusion coefficient making non-linear problem to be more complicated than the linear one. Analyzing the twisted section, an approximately inverse relationship between the proportional factor and the D_0 can be easily derived.

FIG. 5 shows the relationship between D_0 and the reciprocal of the proportional factors fitting by non-linear regression lines. It tells us that, the MDC bias is not only proportional to ξ but also has a reciprocal relationship with D_0 and the scale factor is about $\alpha = -7.1285$. This factor is nearly nothing to do with the experimental parameters as long as the decay curve is complete.

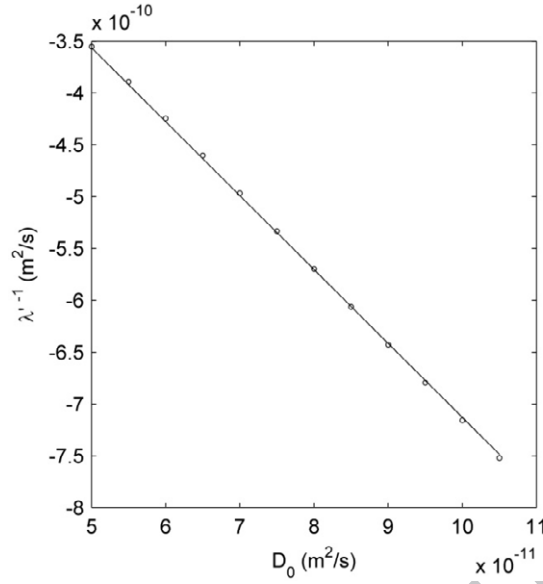


FIG. 5. Relationship between the center diffusion coefficient D_0 and the reciprocal of the proportional factor derived by non-linear regression (open circles) are shown. The solid line represents the regression line of the data. And the fitted line slope is $\alpha = -7.1285$.

Thus, we have the relation of the non-linear form

$$D_{app} - D_0 = \frac{\xi^2}{\alpha D_0}, \quad (16)$$

$$D_0 = \frac{1}{2} \left(D_{app} + \sqrt{D_{app}^2 - \frac{4\xi^2}{\alpha}} \right). \quad (17)$$

Because $4\xi^2/\alpha$ is negative, the whole thing in square root is always positive.

Eq. (17) implies, when $\xi \rightarrow 0$, $D_0 - D_{app} \rightarrow 0$, which is obvious and agrees well with our previous model. Because this scale factor does not depend on any experimental parameters, it can be considered as a constant. Thus, if the ξ is estimated correctly, the right average MDC value could be obtained.

So far we have given an accurate way to calculate the MDC bias from both linear and non-linear regression methods. Our study reveals a systematic error that cannot

be simply eliminated by enhancing only the performances of the instrument. To improve the measurement, some mathematical models based on physical grounds can be applied, e.g. normal distribution as discussed in this paper (Eq. (7), taking A_0 , D_0 and σ as unknowns), stretched/compressed exponential functions,^[23–26] etc.

It should be noted that the MDC bias in this article refers to the systematic deviation between the apparent diffusion coefficient acquired by least squares fitting using Stejskal-Tanner equation and the statistical mean value of the MDC distribution. This bias should not be confused with the biases among the number-average diffusion coefficient, D_N , the weight-average diffusion coefficient, D_W , and the most probable diffusion coefficient, D_P , which are derived from the MDC distribution of polymers.^[10,27] More explicitly, the statistical mean value of the MDC distribution, D_0 , can be considered as the number-average diffusion coefficient of a polydisperse systems. To those who needs to know the Molecular Weight (MW) distribution, some regularization methods, such as Constrained Regularization (CONTIN),^[28] Maximum Entropy (MaxEnt),^[29] Iterative Thresholding Algorithm for Multiexponential Decay (ITAMeD),^[30] Trust-Region Algorithm for the Inversion (TRAIIn),^[31] etc., may be applied together with the use of a scaling law. The advantages and disadvantages among these methods were discussed by J. Jakes, Iain J. Day and K. Xu,^[31–33] which are beyond the scope of this article.

Conclusion

A systematic bias between the apparent diffusion coefficient obtained by the Stejskal–Tanner equation and the statistical mean value of the distribution is noticed and illustrated by numerical analysis and simulations. This systematic bias is caused by improper modeling of the MDC distribution. Under a Gaussian distribution of the MDCs, the bias can be expressed by $D_{app} - D_0 = \lambda^2$.

The MDC obtained by the conventional method is deviated from the statistical mean value (the number-average diffusion coefficient). It requires us to employ appropriate techniques for diffusion data analysis, especially for polydisperse samples. Although our data is derived from a normal distribution, the tendency that the broader the distribution, the larger the bias, remains for most of the distributions. The bias inevitably leads to discrepancies in the calculation of the molecular sizes via the Stokes–Einstein equation. To make the measurement more appropriate, the distribution of MDCs rather than a single MDC value needs to be considered.

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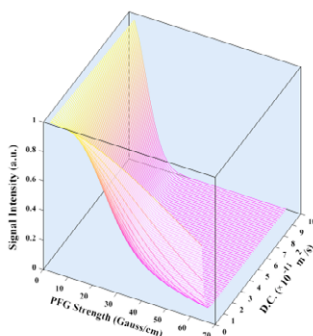
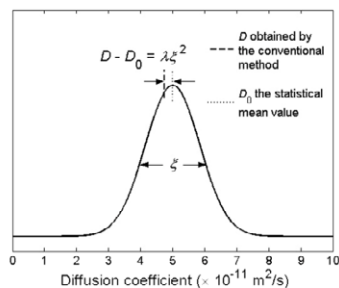
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Graphical abstract



Highlights

- Up to 5% underestimation of the MDC is introduced by fitting data of polydisperse systems.
- The bias is caused by the change of statistical weight in fitting process.
- The bias is increased linearly with the growth of MDC distribution width.